

DAE 2

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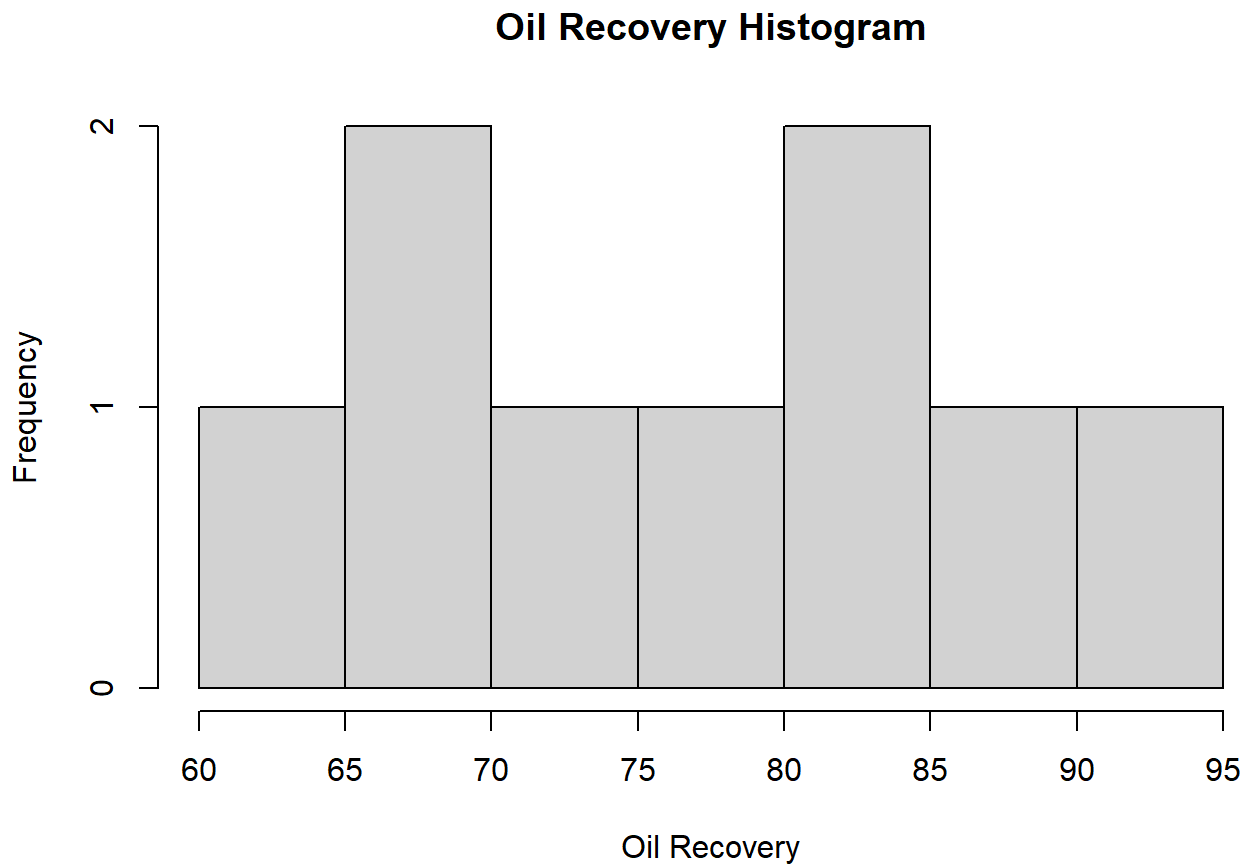
2026-04-11

```
CrudeOil <- read.csv ("CRUDEOIL.csv")  
head(CrudeOil)
```

##	PRESSURE	DIPANGLE	OILREC
## 1	1000	0	60.58
## 2	1000	15	72.72
## 3	1000	30	79.99
## 4	1500	0	66.83
## 5	1500	15	80.78
## 6	1500	30	89.78

Q1

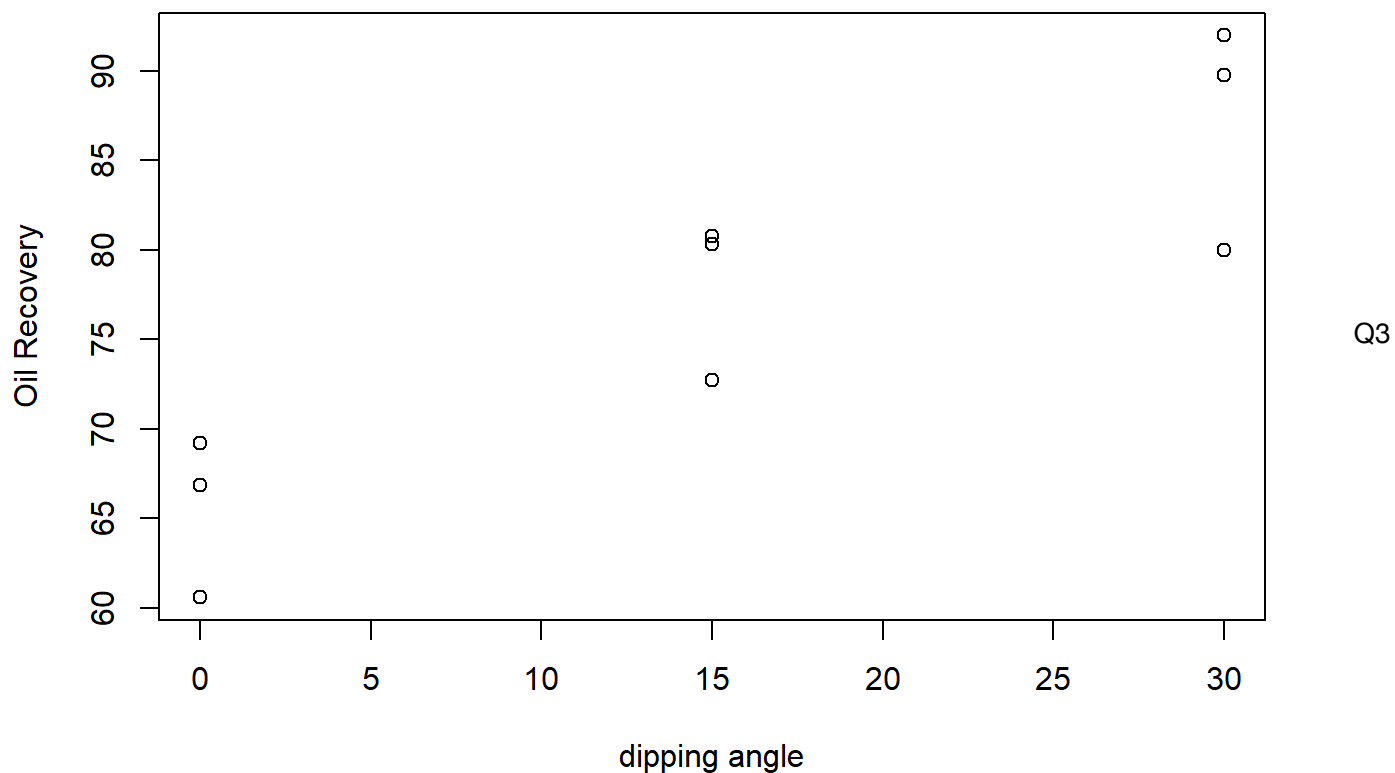
```
hist(CrudeOil$OILREC,  
     main= "Oil Recovery Histogram",  
     xlab= "Oil Recovery")
```



Q2

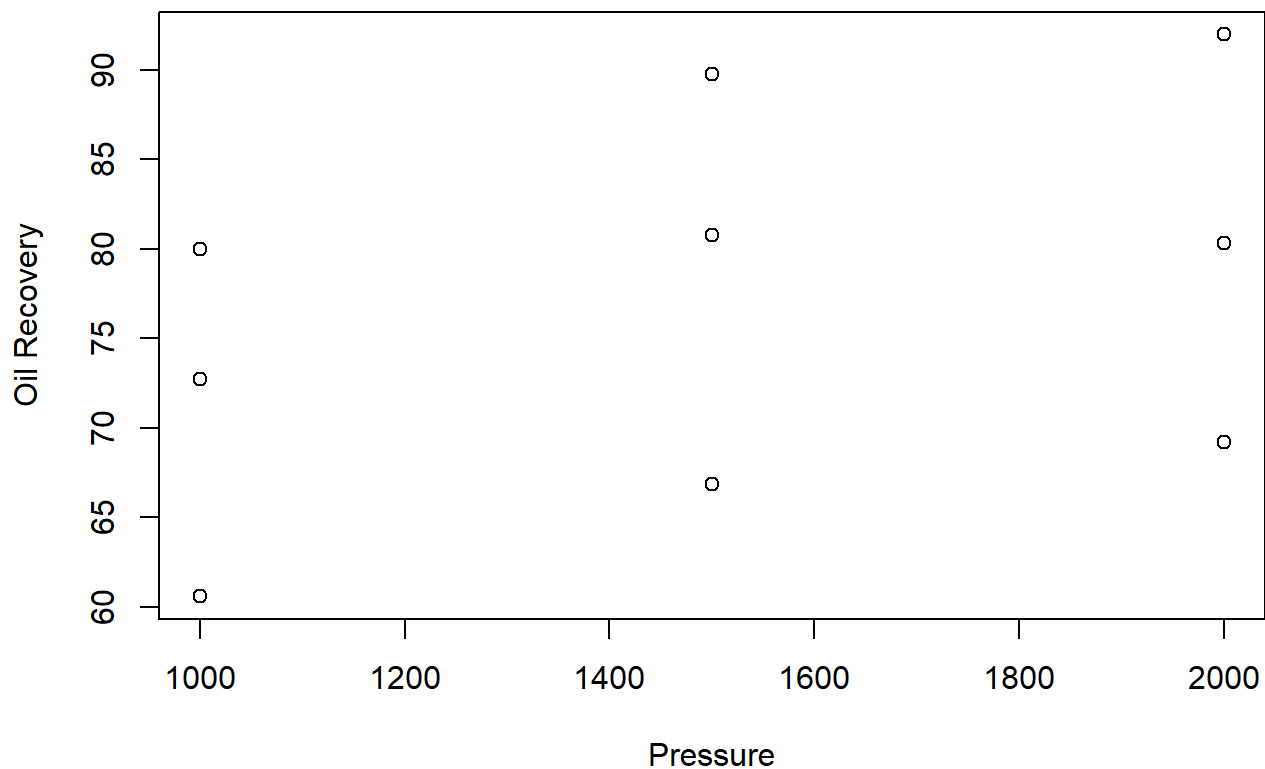
```
plot(CrudeOil$DIPANGLE,CrudeOil$OILREC,  
      xlab= "dipping angle",  
      ylab= "Oil Recovery",  
      main= "Oil Recovery vs Dipping Angle")
```

Oil Recovery vs Dipping Angle



```
plot(CrudeOil$PRESSURE,CrudeOil$OILREC,  
      xlab="Pressure",  
      ylab="Oil Recovery",  
      main= "oil recovery vs pressure")
```

oil recovery vs pressure



Q4

```
CrudeOil$PRESSURE_c <- CrudeOil$PRESSURE - mean(CrudeOil$PRESSURE)
CrudeOil$DIPANGLE_c <- CrudeOil$DIPANGLE - mean(CrudeOil$DIPANGLE)

head(CrudeOil[, c("PRESSURE_c", "DIPANGLE_c")])
```

```
##   PRESSURE_c DIPANGLE_c
## 1      -500        -15
## 2      -500         0
## 3      -500         15
## 4         0        -15
## 5         0         0
## 6         0         15
```

Q5

```
model <- lm(OILREC ~ PRESSURE_c + DIPANGLE_c + I(PRESSURE_c^2) + I(DIPANGLE_c^2) + PRESSURE_c:DIPANGLE_c, data=CrudeOil)
summary(model)
```

```
##
## Call:
## lm(formula = OILREC ~ PRESSURE_c + DIPANGLE_c + I(PRESSURE_c^2) +
##      I(DIPANGLE_c^2) + PRESSURE_c:DIPANGLE_c, data = CrudeOil)
##
## Residuals:
```

	1	2	3	4	5	6	7	8	9
	0.0100	0.5933	-0.6033	-0.9233	0.6200	0.3033	0.9133	-1.2133	0.3000

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.016e+01	9.069e-01	88.388	3.19e-06 ***
PRESSURE_c	9.397e-03	9.935e-04	9.458	0.002505 **
DIPANGLE_c	7.241e-01	3.312e-02	21.866	0.000209 ***
I(PRESSURE_c^2)	-1.334e-05	3.441e-06	-3.876	0.030400 *
I(DIPANGLE_c^2)	-6.867e-03	3.824e-03	-1.796	0.170407
PRESSURE_c:DIPANGLE_c	1.133e-04	8.112e-05	1.397	0.256772

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.217 on 3 degrees of freedom
## Multiple R-squared:  0.9949, Adjusted R-squared:  0.9865
## F-statistic: 117.6 on 5 and 3 DF, p-value: 0.001223
```

Q6

```
summary(model)
```

```
##
## Call:
## lm(formula = OILREC ~ PRESSURE_c + DIPANGLE_c + I(PRESSURE_c^2) +
##      I(DIPANGLE_c^2) + PRESSURE_c:DIPANGLE_c, data = CrudeOil)
##
## Residuals:
##      1      2      3      4      5      6      7      8      9
## 0.0100 0.5933 -0.6033 -0.9233 0.6200 0.3033 0.9133 -1.2133 0.3000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.016e+01  9.069e-01  88.388 3.19e-06 ***
## PRESSURE_c      9.397e-03  9.935e-04   9.458 0.002505 **
## DIPANGLE_c      7.241e-01  3.312e-02  21.866 0.000209 ***
## I(PRESSURE_c^2) -1.334e-05  3.441e-06  -3.876 0.030400 *
## I(DIPANGLE_c^2) -6.867e-03  3.824e-03  -1.796 0.170407
## PRESSURE_c:DIPANGLE_c 1.133e-04  8.112e-05   1.397 0.256772
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.217 on 3 degrees of freedom
## Multiple R-squared:  0.9949, Adjusted R-squared:  0.9865
## F-statistic: 117.6 on 5 and 3 DF, p-value: 0.001223
```

Q7

```
coef(summary(model))["PRESSURE_c:DIPANGLE_c", ]
```

```
##      Estimate  Std. Error    t value    Pr(>|t|)
## 1.133333e-04 8.111689e-05 1.397161e+00 2.567721e-01
```

Q9

```
coef(summary(model))["I(DIPANGLE_c^2)", ]
```

```
##      Estimate  Std. Error    t value    Pr(>|t|)
## -0.006866667 0.003823887 -1.795729451 0.170407335
```

Q10

```
coef(summary(model))["I(PRESSURE_c^2)", ]
```

```
##      Estimate  Std. Error    t value    Pr(>|t|)
## -1.334000e-05 3.441498e-06 -3.876219e+00 3.040032e-02
```

Q13

```
CrudeOil$PRESSURE_c <- CrudeOil$PRESSURE - mean(CrudeOil$PRESSURE)
CrudeOil$DIPANGLE_c <- CrudeOil$DIPANGLE - mean(CrudeOil$DIPANGLE)
```

```
updated_model <- lm(OILREC~ PRESSURE_c + DIPANGLE_c + I(PRESSURE_c^2), data=CrudeOil)
summary(updated_model)
```

```
##
## Call:
## lm(formula = OILREC ~ PRESSURE_c + DIPANGLE_c + I(PRESSURE_c^2),
##     data = CrudeOil)
##
## Residuals:
```

	1	2	3	4	5	6	7	8	9
	0.3450	1.6233	-1.9683	-1.4383	1.6500	-0.2117	-0.4517	-0.1833	0.6350

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.913e+01	8.984e-01	88.084	3.57e-09 ***
PRESSURE_c	9.397e-03	1.270e-03	7.396	0.000711 ***
DIPANGLE_c	7.241e-01	4.235e-02	17.099	1.25e-05 ***
I(PRESSURE_c^2)	-1.334e-05	4.401e-06	-3.031	0.029044 *

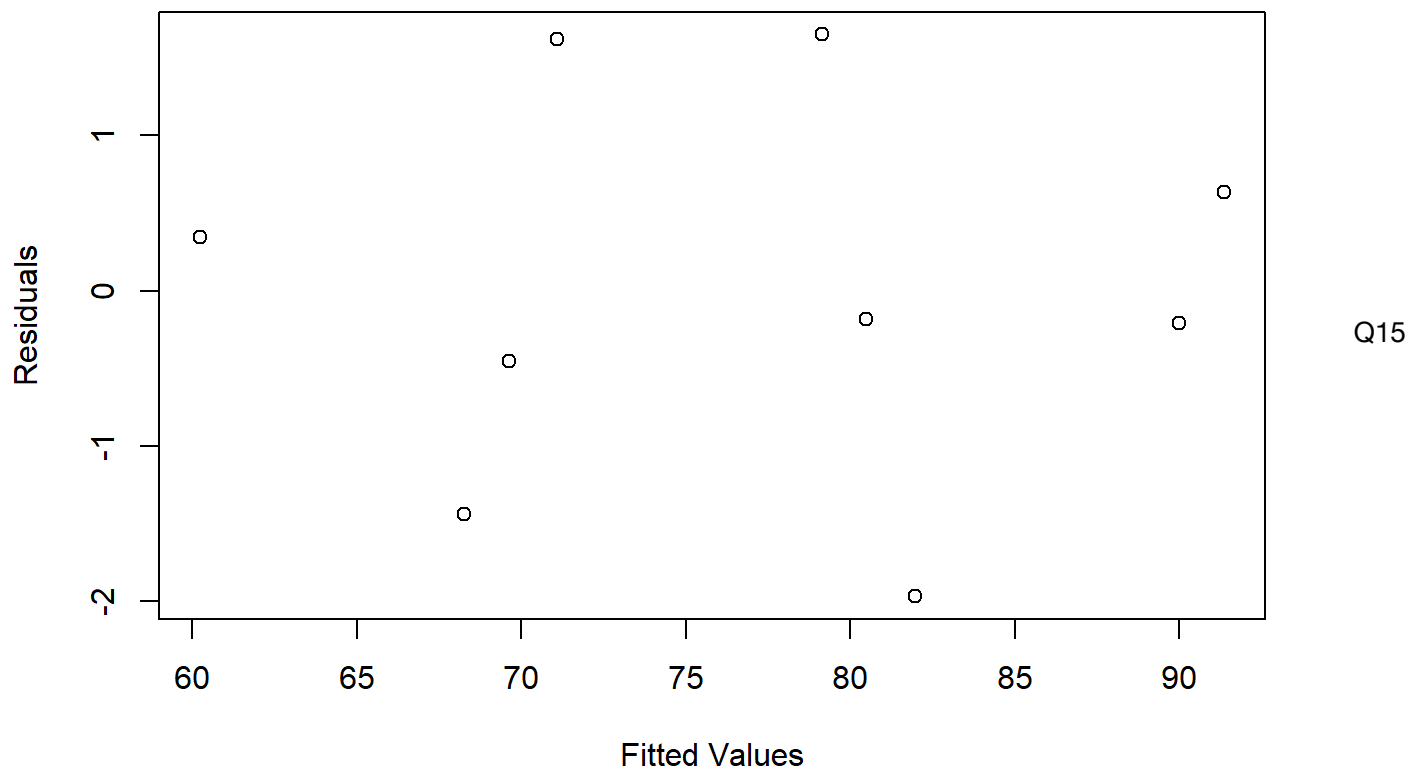
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.556 on 5 degrees of freedom
## Multiple R-squared:  0.9862, Adjusted R-squared:  0.9779
## F-statistic: 118.8 on 3 and 5 DF,  p-value: 4.568e-05
```

```
coef(summary(updated_model))[PRESSURE_c,] coef(summary(updated_model))[DIPANGLE_c,] ##
```

Q14

```
plot(fitted(updated_model), residuals(updated_model),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted Values (Updated Model)")
```

Residuals vs Fitted Values (Updated Model)



```
qqnorm(residuals(updated_model),  
       main = "Normal Q-Q Plot of Residuals (Updated Model)")
```

Normal Q-Q Plot of Residuals (Updated Model)

